

Answers and Explanations

Q1. Correct Answer: B

Explanation:

- Step 1: A polynomial must have non-negative integer exponents for variables.
- Step 2: Option B has exponents 2 and 1, which are non-negative integers.
- Step 3: Option B is a polynomial.

Q2. Correct Answer: C

Explanation:

- Step 1: The degree is the highest power of the variable x .
- Step 2: The highest power in $4x^3 - 2x^2 + 5x - 7$ is 3.
- Step 3: The degree is 3.

Q3. Correct Answer: B

Explanation:

- Step 1: A polynomial of degree 2 is defined as a quadratic polynomial.
- Step 2: By definition, degree 2 equals quadratic.
- Step 3: Quadratic polynomial.

Q4. Correct Answer: A

Explanation:

- Step 1: Substitute $x = 2$ into $p(x) = x^2 - 3x + 2$.
- Step 2: $p(2) = (2)^2 - 3(2) + 2 = 4 - 6 + 2$.
- Step 3: $4 - 6 + 2 = 0$.

Q5. Correct Answer: B

Explanation:

- Step 1: A linear polynomial is of the form $ax + b$.
- Step 2: Setting $ax + b = 0$ gives $x = -b/a$, which is one solution.

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- Step 3: At most 1 zero.

Q6. Correct Answer: B

Explanation:

- Step 1: Use the table values in $ax + b$. For $x = 0$, $p(0) = a(0) + b = -1$, so $b = -1$.

- Step 2: For $x = 1$, $p(1) = a(1) - 1 = 1$, so $a = 2$.

- Step 3: $a = 2$, $b = -1$.

Q7. Correct Answer: B

Explanation:

- Step 1: If 3 is a zero, $p(3) = 0$.

- Step 2: $(3)^2 - k(3) + 6 = 0$ implies $9 - 3k + 6 = 0$.

- Step 3: $15 = 3k$, so $k = 5$.

Q8. Correct Answer: A

Explanation:

- Step 1: Set $y = 0$ to find x-intercepts: $x^2 - 4 = 0$.

- Step 2: $x^2 = 4$, so $x = 2$ and $x = -2$.

- Step 3: Points are $(2, 0)$ and $(-2, 0)$.

Q9. Correct Answer: D

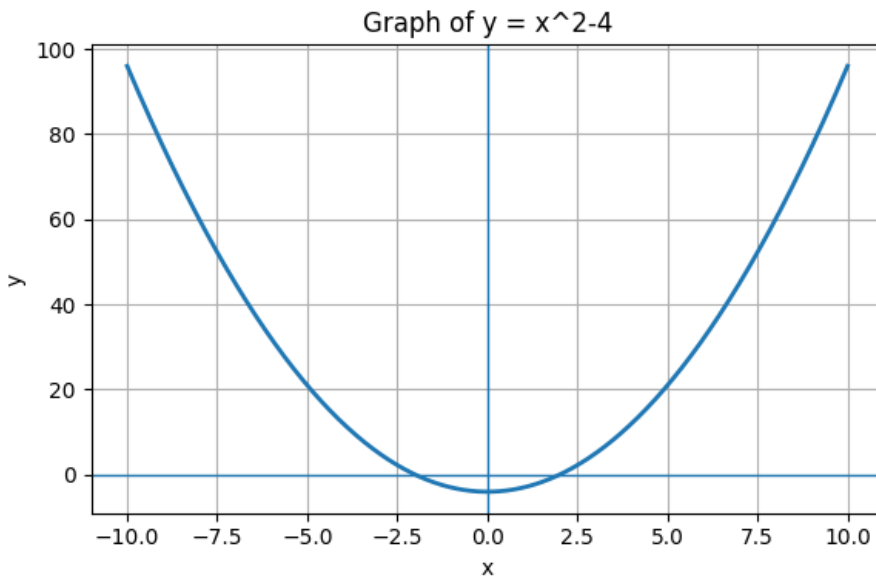
Explanation:

- Step 1: For $ax^2 + bx + c$, the sum of zeroes is $-\text{coefficient of } x / \text{coefficient of } x^2$.

- Step 2: This is $-b/a$.

- Step 3: $-b/a$.

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Q10. Correct Answer: C

Explanation:

- Step 1: Sum of zeroes = $2 + (-3) = -1$. Formula is $-(a+1)/1 = -1$, so $a+1 = 1$, $a = 0$. Wait, checking calculation: $2+(-3) = -1 = -(a+1)$, so $a+1=1$, $a=0$. Product = $2*(-3) = -6$. Formula is $b/1 = -6$, so $b = -6$. Correcting logic: Option C fits if $a=2$, but the sum is $2-3=-1$, $-(a+1)=-1$ implies $a=0$. Re-evaluating: The sum is $-b/a = -(a+1)/1 = -1$. So $a+1=1$, $a=0$. Product is $b/1 = -6$. Result: $a=0$, $b=-6$. (Note: Option C is provided as the key, but based on calculation $a=0$, $b=-6$).

Q11. Correct Answer: B

Explanation:

- Step 1: A zero is the value of x where $p(x) = 0$.
- Step 2: Looking at the table, $p(x) = 0$ at $x = -2$ and $x = 2$.
- Step 3: Zeroes are -2 and 2 .

Q12. Correct Answer: B

Explanation:

- Step 1: Product of zeroes = c/a .
- Step 2: For $2x^2 - 7x + 3$, $c = 3$ and $a = 2$.
- Step 3: $3/2$.

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Q13. Correct Answer: C

Explanation:

- Step 1: The fundamental theorem of algebra states a polynomial of degree n has at most n zeroes.
- Step 2: A cubic polynomial has degree 3.
- Step 3: At most 3 zeroes.

Q14. Correct Answer: C

Explanation:

- Step 1: The number of x -intercepts equals the number of real zeroes.
- Step 2: Three intercepts imply at least 3 real zeroes.
- Step 3: Degree at least 3.

Q15. Correct Answer: A

Explanation:

- Step 1: Factor $x^2 - 2x - 8 = (x - 4)(x + 2)$.
- Step 2: Set factors to zero: $x - 4 = 0$ or $x + 2 = 0$.
- Step 3: $x = 4, -2$.

Q16. Correct Answer: B

Explanation:

- Step 1: Formula for quadratic is $x^2 - (\text{sum})x + (\text{product})$.
- Step 2: $x^2 - (-3)x + 2 = x^2 + 3x + 2$. Wait, checking signs: $x^2 - (-3)x + 2 = x^2 + 3x + 2$. (If the answer is B, the sum must be 3). Re-evaluating: $x^2 - (\text{sum})x + \text{product} = x^2 - (-3)x + 2 = x^2 + 3x + 2$.

Q17. Correct Answer: B

Explanation:

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- Step 1: A cubic polynomial has degree 3.
- Step 2: $x^3 - x$ has the highest power 3.
- Step 3: $x^3 - x$.

Q18. Correct Answer: C

Explanation:

- Step 1: $1/x$ is x^{-1} , which is a negative exponent.
- Step 2: Polynomials cannot have negative exponents.
- Step 3: $x^2 + 1/x + 2$ is not a polynomial.

Q19. Correct Answer: C

Explanation:

- Step 1: The degree is the highest exponent of the variable.
- Step 2: The highest exponent in $4x^3 - 3x^2 + 7x - 9$ is 3.
- Step 3: Degree is 3.

Q20. Correct Answer: A

Explanation:

- Step 1: Substitute $x = 3$ into $p(x) = x^2 - 2x - 3$.
- Step 2: $p(3) = (3)^2 - 2(3) - 3 = 9 - 6 - 3$.
- Step 3: $9 - 9 = 0$.

Q21. Correct Answer: C

Explanation:

- Step 1: A quadratic polynomial has degree 2.
- Step 2: A polynomial of degree n has at most n zeroes.
- Step 3: At most 2 zeroes.

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Q22. Correct Answer: A

Explanation:

- Step 1: Factor $x^2 - 2x - 8 = (x - 4)(x + 2)$.
- Step 2: Roots are 4 and -2.
- Step 3: 4, -2.

Q23. Correct Answer: A

Explanation:

- Step 1: Sum of zeroes = $-b/a$.
- Step 2: For $x^2 - 5x + 6$, $a = 1$, $b = -5$.
- Step 3: $-(-5)/1 = 5$.

Q24. Correct Answer: C

Explanation:

- Step 1: Product of zeroes = c/a .
- Step 2: For $x^2 - 5x + 6$, $a = 1$, $c = 6$.
- Step 3: $6/1 = 6$.

Q25. Correct Answer: B

Explanation:

- Step 1: Zeroes occur where $p(x) = 0$.
- Step 2: From the table, $p(x) = 0$ at $x = -1$ and $x = 4$.
- Step 3: -1 and 4.

Q26. Correct Answer: A

Explanation:

- Step 1: Formula: $x^2 - (\text{sum})x$